

Dockerized-Microservices — Cheat Sheet

Chain C · Project C-2 · dense one-page recap — pairs with docs/interactive/index.html

1 · Dockerfile Layers

- Order: `COPY requirements.txt` → `pip install` → `COPY app/`
- Each instruction = one cached layer; cache busts from first change onward
- Multi-stage: `builder` stage has compilers, final stage only has the installed packages

2 · Compose + Celery

Concept	What it does
<code>depends_on: service_healthy</code>	Waits for healthcheck, not just container start
Service-name DNS	<code>redis://redis:6379</code> resolves via Docker's embedded DNS
Same image, 2 roles	<code>api</code> and <code>worker</code> share one <code>build:</code> , differ by <code>command:</code>

- **Bug to remember:** Celery needs `include=["app.tasks"]` or the worker never registers the task — fails silently, no error anywhere

Service topology

Service	Role	Exposed?
<code>redis</code>	Celery broker + result backend	No — internal only
<code>api</code>	FastAPI, enqueues tasks via <code>.delay()</code>	No — internal only
<code>worker</code>	Same image as <code>api</code> , runs as Celery consumer	No — internal only
<code>nginx</code>	Reverse proxy, static file server	Yes — host:8080

3 · Nginx Reverse Proxy

Location block	Routes to
<code>/api/</code>	<code>proxy_pass http://api:8000/</code>
<code>/</code>	Static files, <code>try_files \$uri /index.html</code>

- Only `nginx` publishes a host port — `api` / `redis` stay internal

4 · CI/CD

Job	Gate
<code>test</code>	<code>pytest</code> — must pass
<code>build-and-push</code>	<code>needs: test</code> — skipped if test fails

- GHCR auth uses auto-provided `secrets.GITHUB_TOKEN` — no manual setup
- Image tagged by git SHA: `ghcr.io/<repo>:<sha>`